Quantum Gate Operations.

Hadamard Gate: $\frac{1}{\sqrt{2}}\begin{bmatrix}1 & 1\\ 1 & -1\end{bmatrix}$

$$H|0\rangle = \frac{1}{\sqrt{2}}\begin{bmatrix}1 & 1\\ 1 & -1\end{bmatrix}\begin{bmatrix}1\\ 0\end{bmatrix} = \frac{1}{\sqrt{2}}\begin{bmatrix}1+0\\ 1-0\end{bmatrix} = \frac{1}{\sqrt{2}}\begin{bmatrix}1\\ 1\end{bmatrix}$$

2×2 2×1 2×1

$$= \frac{1}{\sqrt{2}}\begin{bmatrix}1\\ 0\end{bmatrix} + \frac{1}{\sqrt{2}}\begin{bmatrix}0\\ 1\end{bmatrix}$$

$$= \frac{1}{\sqrt{2}}|0\rangle + \frac{1}{\sqrt{2}}|1\rangle$$

$H|0\rangle \rightarrow$ $H|1\rangle$

$$H|+\rangle = \frac{1}{\sqrt{2}}\begin{bmatrix}1 & 1\\ 1 & -1\end{bmatrix}\left(\frac{1}{\sqrt{2}}|0\rangle + \frac{1}{\sqrt{2}}|1\rangle\right)$$

$$= \frac{1}{2}\begin{bmatrix}1 & 1\\ 1 & -1\end{bmatrix}\begin{bmatrix}1\\ 0\end{bmatrix} + \quad \cdots$$

$$H|+\rangle = H\left(\frac{1}{\sqrt{2}}|0\rangle + \frac{1}{\sqrt{2}}|1\rangle\right)$$

$$= \frac{1}{\sqrt{2}}H|0\rangle + \frac{1}{\sqrt{2}}H|1\rangle$$

$$= \frac{1}{\sqrt{2}}\left[|+\rangle + |-\rangle\right]$$

$$= \frac{1}{\sqrt{2}}\left[\frac{1}{\sqrt{2}}|0\rangle + \cancel{\frac{1}{\sqrt{2}}|1\rangle} + \frac{1}{\sqrt{2}}|0\rangle - \cancel{\frac{1}{\sqrt{2}}|1\rangle}\right]$$

$$= \frac{1}{\sqrt{2}}\left[\frac{\cancel{2}^{\sqrt{2}}}{\sqrt{2}}|0\rangle\right] = \frac{\cancel{\sqrt{2}}}{\cancel{\sqrt{2}}}|0\rangle = |0\rangle$$

2 Qubit Quantum Gates.

Representation:

$$|\psi\rangle = |q_1\rangle|q_2\rangle = |q_1 q_2\rangle$$

2-Qubit System – basis states

$|00\rangle$, $|01\rangle$, $|10\rangle$, $|11\rangle$

A general two-qubit state can be expressed as,

$$|\psi\rangle = a_{00}|00\rangle + a_{01}|01\rangle + a_{10}|10\rangle + a_{11}|11\rangle$$

ie $|a_{00}|^2 + |a_{01}|^2 + |a_{10}|^2 + |a_{11}|^2 = 1.$

$$|\psi\rangle = \begin{bmatrix}a_{00}\\ a_{01}\\ a_{10}\\ a_{11}\end{bmatrix} \cong a_{00}|00\rangle + a_{01}|01\rangle + a_{10}|10\rangle + a_{11}|11\rangle$$

$$|00\rangle = \begin{bmatrix}\\ \\ \\ \end{bmatrix}$$

Tensor Product $|0\rangle \otimes |0\rangle = \begin{bmatrix}1\\ 0\end{bmatrix} \otimes \begin{bmatrix}1\\ 0\end{bmatrix} = \begin{bmatrix}1\begin{bmatrix}1\\ 0\end{bmatrix}\\ 0\begin{bmatrix}1\\ 0\end{bmatrix}\end{bmatrix} = \begin{bmatrix}1\\ 0\\ 0\\ 0\end{bmatrix}_{4\times 1}$

Outer Product. $|0\rangle\langle 0| = \begin{bmatrix}1\\ 0\end{bmatrix}_{2\times 1} \otimes \begin{bmatrix}1 & 0\end{bmatrix}_{1\times 2}$

$$= \begin{bmatrix}\\ \\ \end{bmatrix}_{2\times 2}$$

Inner Product $\langle 0|0\rangle = \begin{bmatrix}1 & 0\end{bmatrix}_{1\times 2} \otimes \begin{bmatrix}1\\ 0\end{bmatrix}_{2\times 1} = \begin{bmatrix}\ \end{bmatrix}_{1\times 1}$

$$|0\rangle \otimes |1\rangle = |01\rangle = \begin{bmatrix}1\\ 0\end{bmatrix} \otimes \begin{bmatrix}0\\ 1\end{bmatrix} = \begin{bmatrix}1\begin{bmatrix}0\\ 1\end{bmatrix}\\ 0\begin{bmatrix}0\\ 1\end{bmatrix}\end{bmatrix} = \begin{bmatrix}0\\ 1\\ 0\\ 0\end{bmatrix}$$

$$|1\rangle \otimes |0\rangle = \begin{bmatrix}0\\ 1\end{bmatrix} \otimes \begin{bmatrix}1\\ 0\end{bmatrix} = \begin{bmatrix}0\begin{bmatrix}1\\ 0\end{bmatrix}\\ 1\begin{bmatrix}1\\ 0\end{bmatrix}\end{bmatrix} = \begin{bmatrix}0\\ 0\\ 1\\ 0\end{bmatrix}$$

$|11\rangle = ?$

$|00\rangle = \begin{bmatrix}1\\ 0\\ 0\\ 0\end{bmatrix}$ $|01\rangle = \begin{bmatrix}0\\ 1\\ 0\\ 0\end{bmatrix}$ $|10\rangle = \begin{bmatrix}0\\ 0\\ 1\\ 0\end{bmatrix}$ $|11\rangle = \begin{bmatrix}0\\ 0\\ 0\\ 1\end{bmatrix}$

2 Qubit Quantum Gate Operations.

$$CNOT = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{bmatrix} = \begin{bmatrix} I & 0 \\ 0 & X \end{bmatrix}$$

$$|+\rangle = \frac{1}{\sqrt{2}}|0\rangle + \frac{1}{\sqrt{2}}|1\rangle$$

$$|+\rangle|+\rangle = \left(\frac{1}{\sqrt{2}}|0\rangle + \frac{1}{\sqrt{2}}|1\rangle\right) \otimes \left(\frac{1}{\sqrt{2}}|0\rangle + \frac{1}{\sqrt{2}}|1\rangle\right)$$

$$= \frac{1}{2}|00\rangle + \frac{1}{2}|01\rangle + \frac{1}{2}|10\rangle + \frac{1}{2}|11\rangle$$

$$\begin{bmatrix} \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \end{bmatrix} \otimes \begin{bmatrix} \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \end{bmatrix} = \begin{bmatrix} \frac{1}{2} \\ \frac{1}{2} \\ \frac{1}{2} \\ \frac{1}{2} \end{bmatrix} = \frac{1}{2}|00\rangle + \frac{1}{2}|01\rangle + \frac{1}{2}|10\rangle + \frac{1}{2}|11\rangle$$

$$|+\rangle|+\rangle = \frac{1}{2}\left(|00\rangle + |01\rangle + |10\rangle + |11\rangle\right)$$

$$CNOT\,|+\rangle|+\rangle = \frac{1}{2}\left(|00\rangle + |01\rangle + |11\rangle + |10\rangle\right)$$

$$= |+\rangle|+\rangle$$

Try. Represent $|+\rangle, |-\rangle$ in 2 qubit basis state form and then compute

$CNOT\,(|+\rangle|-\rangle)$

Similarly for $|-\rangle|+\rangle$ & $|-\rangle|-\rangle$